SOFTWARE REQUIREMENTS SPECIFICATION

Health Insurance Claim Portal

Document Type: Software Requirements Specification (SRS)
Version: 1.0
Project: Health Insurance Claim Portal
Department: CSE (SEC-B)
Institution: KIET Deemed to be University
Academic Year: 2026–2027
Submitted By: Gagan Sharma
Roll No: 202401100200140
Date: 14/09/2026

Document Revision History

Version	Date	Author	Description
0.1	28/08/2026	Gagan Sharma	Initial draft
0.2	03/09/2026	Gagan Sharma	Functional requirements added
0.3	09/09/2026	Gagan Sharma	System models and non-functional requirements added
1.0	14/09/2026	Gagan Sharma	Final SRS

Table of Contents

1. Introduction
2. Overall Description
3. System Actors and User Classes
4. Functional Requirements
5. External Interface Requirements
6. Non-Functional Requirements
7. System Workflow
8. System Models and Diagrams
9. Data Requirements
10. Business Rules
11. Security and Privacy Requirements

12. Error Handling and Exception Requirements
13. Acceptance Criteria
14. Testing Requirements
15. Requirements Traceability Matrix
16. System Constraints
17. Assumptions and Dependencies
18. Limitations
19. Future Enhancements
20. Conclusion
21. Appendices

---

1. Introduction

1.1 Purpose

This document lays out the functional and non-functional requirements for the **Health Insurance Claim Portal**, a project undertaken as part of the CSE (SEC-B) coursework at KIET.

Health insurance claims are usually handled through paperwork and manual follow-ups, which makes the process slow for both the policyholder and the insurer. The idea behind this project is to move that process onto a single web platform, where a policyholder can submit a claim along with the required documents, keep track of where it stands, and eventually see whether it was approved and what settlement amount was decided.

On the other side, the same platform should let claim officers and assessors check the information a policyholder has submitted, look at the supporting documents, and take a decision - approve, reject, or ask for more documents - without needing a separate system for that.

The rest of this document tries to capture that behavior in detail - what the system should do, what it should not do, how it should behave under different conditions, and what constraints and quality expectations apply to it.

1.2 Scope

This project is limited to a web-based application - it does not try to cover mobile apps or any physical/offline part of claim handling. The scope, for this version, is to digitize the claim-processing workflow end to end.

That means starting from policyholder login and claim submission, going through document verification and assessment by the insurance side, and ending at approval/rejection, settlement, and closure of the claim.

In Scope

The proposed system includes:

- User registration and authentication
- Role-based access control
- Policyholder profile management
- Insurance policy information
- Claim creation and submission
- Claim information validation
- Supporting document upload
- Claim verification
- Claim assessment
- Claim approval and rejection
- Claim status tracking
- Claim history
- Settlement information
- Notifications regarding claim status
- Administrative management
- Reporting and monitoring
- Secure storage and retrieval of claim-related information
- Audit logging of important claim-processing activities

Out of Scope

Unless explicitly implemented as part of the project, the following are outside the scope:

- Medical diagnosis or treatment
- Hospital management
- Insurance underwriting
- Direct medical consultation
- Physical delivery of documents
- Automated medical decision-making
- Direct integration with government health databases
- Real-world banking/payment settlement infrastructure
- Insurance policy creation through external insurance providers

---

1.3 Intended Audience

This document is intended for:

- Project developers
- Project supervisors
- Software testers
- System administrators
- Insurance personnel
- Academic evaluators
- Future maintainers of the system
- Other stakeholders involved in the project

1.4 Definitions

Term	Definition
SRS	Software Requirements Specification
Policyholder	Person who owns or is covered by an insurance policy
Claim	Request submitted by a policyholder for insurance coverage or reimbursement
Claim Officer	Authorized employee responsible for reviewing and processing claims
Claim Assessor	Authorized personnel responsible for evaluating claim eligibility
Administrator	User responsible for managing the system
Policy	Insurance contract containing coverage and applicable conditions
Settlement	Amount determined to be payable for an approved claim
Claim ID	Unique identifier assigned to a claim
RBAC	Role-Based Access Control
API	Application Programming Interface
UI	User Interface
DB	Database

1.5 References

The project documentation may refer to:

- IEEE software requirements engineering practices

- Relevant software engineering principles
- Database design principles
- Web application security practices
- Documentation of technologies used by the project
- Relevant insurance-domain requirements defined for the project

---

1.6 Document Overview

This document begins with an overall description of the Health Insurance Claim Portal, followed by user roles, functional requirements, interfaces, non-functional requirements, workflows, system models, data requirements, business rules, security requirements, testing requirements, limitations, and future enhancements.

---

2. Overall Description

2.1 Product Perspective

This portal is meant to run as a single, centralized web application rather than a set of separate tools for policyholders and insurance staff.

Both sides use the same platform - policyholders to manage their claims, and claim officers/assessors/admins to process them - just through different views based on their role.

A rough architecture diagram for this would look like:

User Interface → Application/API Layer → Business Logic → Database/Storage

The application layer handles authentication, authorization, claim processing, validation, business rules, document management, and communication with persistent storage.

2.2 Major Product Functions

The major functions of the system are:

1. User authentication
2. User and role management
3. Policy management
4. Claim submission
5. Document management
6. Claim verification
7. Claim assessment
8. Claim approval

9. Claim rejection
10. Claim settlement
11. Claim tracking
12. Notifications
13. Administrative management
14. Reporting
15. Audit logging

---

3. System Actors and User Classes

3.1 Policyholder

The policyholder is essentially the end customer using this portal - the person whose claim is going through the system.

Here's what a policyholder should be able to do on the portal:

- Register an account
- Log into the system
- View profile information
- View policy information
- Submit a claim
- Upload supporting documents
- View submitted claims
- Track claim status
- View claim history
- View approval/rejection information
- View settlement information
- Receive claim-related notifications

3.2 Claim Officer

Next is the claim officer, who works on the insurer's side and handles the first-level processing of a claim once it comes in.

Their responsibilities on the portal include being able to:

- Log into the system
- View assigned or available claims

- Review claim details
- Review supporting documents
- Verify claim information
- Request additional documents
- Update claim-processing status
- Forward claims for assessment
- Record remarks

3.3 Claim Assessor

Once a claim officer has verified a claim, it moves to the claim assessor, whose job is to check it against the actual policy conditions and decide whether it is eligible.

On the portal, this role is expected to:

- Review verified claims
- Examine policy coverage information
- Review supporting documents
- Evaluate claim eligibility
- Record assessment decisions
- Recommend or approve settlement amounts
- Approve eligible claims
- Reject ineligible claims with reasons

3.4 Administrator

Finally, the administrator sits above these roles and is responsible for keeping the application running smoothly overall.

This role is expected to:

- Manage users
- Manage roles
- Manage insurance policy information
- Manage insurance employees
- Monitor claims
- View system activity
- Generate reports

- Manage system configuration
- Review audit records

---

4. Functional Requirements

Functional requirements define what the system shall do.

4.1 Authentication and Authorization

FR-AUTH-001 — User Registration

The system shall allow eligible users to create an account by providing the required registration information.

FR-AUTH-002 — User Login

The system shall authenticate registered users using valid credentials.

FR-AUTH-003 — Invalid Login Handling

The system shall reject invalid authentication attempts and display an appropriate error message.

FR-AUTH-004 — Role-Based Access

The system shall provide access to system functionality according to the authenticated user's assigned role.

FR-AUTH-005 — Protected Resources

The system shall prevent unauthenticated users from accessing protected resources.

FR-AUTH-006 — Logout

The system shall allow authenticated users to terminate their session securely.

FR-AUTH-007 — Credential Protection

The system shall protect authentication credentials using appropriate security mechanisms.

4.2 Policyholder Management

FR-POLH-001 — View Profile

The system shall allow policyholders to view their profile information.

FR-POLH-002 — Update Profile

The system shall allow policyholders to update information that they are authorized to modify.

FR-POLH-003 — View Policy Information

The system shall allow policyholders to view their associated insurance policy information.

FR-POLH-004 — Claim History

The system shall allow policyholders to view their previous claims.

4.3 Policy Management

FR-POL-001 — Policy Information

The system shall maintain relevant insurance policy information required for claim processing.

FR-POL-002 — Policy Status

The system shall maintain the status of each policy.

Possible policy statuses include:

- Active
- Expired
- Suspended
- Cancelled

FR-POL-003 — Policy Verification

The system shall allow authorized personnel to verify the policy associated with a claim.

FR-POL-004 — Coverage Information

The system shall provide authorized users with applicable policy coverage information required for claim assessment.

4.4 Claim Submission

FR-CLM-001 — Create Claim

The system shall allow an eligible policyholder to initiate a health insurance claim.

FR-CLM-002 — Enter Claim Information

The system shall allow the policyholder to provide the information required to process the claim.

Potential claim information includes:

- Policy number
- Patient information
- Hospital information
- Treatment information
- Admission date
- Discharge date
- Claim amount

- Description of treatment
- Supporting information

FR-CLM-003 — Mandatory Field Validation

The system shall validate mandatory claim fields before submission.

FR-CLM-004 — Claim Identifier

The system shall assign a unique claim identifier to each successfully submitted claim.

FR-CLM-005 — Claim Submission Confirmation

The system shall provide confirmation after successful claim submission.

FR-CLM-006 — Duplicate Claim Prevention

The system should identify potential duplicate claims according to configured business rules.

4.5 Document Management

FR-DOC-001 — Upload Documents

The system shall allow policyholders to upload supporting documents associated with a claim.

Examples include:

- Hospital bills
- Medical reports
- Discharge summaries
- Prescriptions
- Diagnostic reports
- Identity documents

FR-DOC-002 — Document Validation

The system shall validate uploaded files according to configured file-type and file-size restrictions.

FR-DOC-003 — Claim Association

The system shall associate each uploaded document with its corresponding claim.

FR-DOC-004 — Document Retrieval

Authorized personnel shall be able to retrieve documents associated with claims they are authorized to process.

FR-DOC-005 — Unauthorized Document Access Prevention

The system shall prevent unauthorized users from accessing claim documents.

4.6 Claim Verification

FR-VER-001 — Review Claim

Authorized claim officers shall be able to review submitted claim information.

FR-VER-002 — Review Documents

Authorized claim officers shall be able to review supporting documents.

FR-VER-003 — Verification Status

The system shall maintain the verification status of a claim.

FR-VER-004 — Additional Documentation

The system shall allow authorized personnel to request additional documentation when required.

FR-VER-005 — Verification Remarks

The system shall allow authorized personnel to record verification remarks.

4.7 Claim Assessment

FR-ASS-001 — Assess Claim

The system shall allow authorized claim assessors to evaluate verified claims.

FR-ASS-002 — Policy-Based Evaluation

The system shall provide relevant policy and claim information required for claim assessment.

FR-ASS-003 — Assessment Decision

The system shall allow authorized assessors to record an assessment decision.

FR-ASS-004 — Assessment Remarks

The system shall allow assessors to record relevant remarks associated with an assessment.

4.8 Claim Approval

FR-APP-001 — Approve Claim

Authorized personnel shall be able to approve eligible claims.

FR-APP-002 — Approval Record

The system shall record the approval decision and relevant timestamp.

FR-APP-003 — Approval Status

The system shall update the claim status after approval.

4.9 Claim Rejection

FR-REJ-001 — Reject Claim

Authorized personnel shall be able to reject claims that do not satisfy applicable requirements.

FR-REJ-002 — Rejection Reason

The system shall require an appropriate reason when rejecting a claim.

FR-REJ-003 — Rejection Record

The system shall store the rejection decision, reason, responsible user, and relevant timestamp.

FR-REJ-004 — Rejection Notification

The system shall notify the policyholder when a claim is rejected, where notification functionality is enabled.

4.10 Claim Settlement

FR-SET-001 — Settlement Amount

The system shall allow authorized personnel to record the approved settlement amount.

FR-SET-002 — Settlement Information

The system shall maintain relevant settlement information associated with an approved claim.

FR-SET-003 — Settlement Status

The system shall maintain the settlement status of an approved claim.

FR-SET-004 — Claim Closure

The system shall allow an appropriately authorized user or process to mark a settled claim as closed.

4.11 Claim Tracking

FR-TRK-001 — Current Claim Status

The system shall allow policyholders to view the current status of their claims.

FR-TRK-002 — Claim Timeline

The system should provide a chronological representation of significant claim-processing events where supported.

FR-TRK-003 — Claim Details

The system shall allow authorized users to view relevant claim information.

FR-TRK-004 — Claim History

The system shall maintain and display previous claim records to authorized policyholders and employees.

4.12 Notifications

FR-NOT-001 — Claim Submission Notification

The system should notify the policyholder after successful claim submission.

FR-NOT-002 — Status Change Notification

The system should notify the policyholder when a claim status changes.

FR-NOT-003 — Additional Documents Notification

The system should notify the policyholder when additional documents are required.

FR-NOT-004 — Decision Notification

The system should notify the policyholder when the claim is approved or rejected.

4.13 Administration

FR-ADM-001 — User Management

The administrator shall be able to view and manage registered users.

FR-ADM-002 — Employee Management

The administrator shall be able to manage authorized insurance personnel.

FR-ADM-003 — Claim Monitoring

The administrator shall be able to monitor claims processed through the system.

FR-ADM-004 — System Reports

The administrator should be able to generate relevant system and claim reports.

FR-ADM-005 — Audit Monitoring

The administrator shall be able to review important system activities where audit functionality is implemented.

---

5. External Interface Requirements

5.1 User Interface

The system shall provide an intuitive and consistent web interface.

The interface should include:

Policyholder Interface

- Login
- Registration
- Dashboard

- Profile
- Policy details
- Submit claim
- Document upload
- Claim list
- Claim details
- Claim tracking
- Settlement information

Employee Interface

- Employee dashboard
- Claim queue
- Claim details
- Document viewer
- Verification interface
- Assessment interface
- Approval/rejection interface

Administrator Interface

- Administrative dashboard
- User management
- Employee management
- Policy management
- Claim monitoring
- Reports
- Audit information

5.2 Software Interface

The system may interact with:

- Database management system
- File/document storage
- Authentication mechanisms
- Notification services

- External APIs, where applicable

The exact interfaces shall depend on the technologies used in the implementation.

5.3 Hardware Interface

The system shall operate on standard computing devices capable of running a supported web browser.

No specialized hardware shall be required for the basic operation of the portal unless explicitly added as a future feature.

5.4 Communication Interface

The application shall use standard network communication protocols for communication between the client, server, and external services.

Where sensitive information is transmitted, secure communication shall be used.

---

6. Non-Functional Requirements

6.1 Performance

NFR-PERF-001

The system should provide normal application responses within an acceptable response time under expected operating conditions.

NFR-PERF-002

The system should efficiently process claim searches and retrieval operations.

NFR-PERF-003

The system should support multiple concurrent users according to the capacity of the deployed infrastructure.

6.2 Security

NFR-SEC-001

The system shall implement authentication for protected resources.

NFR-SEC-002

The system shall implement role-based authorization.

NFR-SEC-003

The system shall prevent unauthorized users from accessing claims and documents.

NFR-SEC-004

Passwords shall not be stored in plaintext.

NFR-SEC-005

The system shall validate user input to reduce security vulnerabilities.

NFR-SEC-006

The system shall validate uploaded files.

NFR-SEC-007

The system shall protect sensitive information during transmission.

NFR-SEC-008

The system shall prevent users from accessing resources belonging to other users without appropriate authorization.

6.3 Privacy

Because the system processes personal, insurance, and potentially medical information, privacy shall be treated as a major system requirement.

The system shall:

- Restrict sensitive information to authorized users.
- Avoid unnecessary exposure of personal information.
- Protect uploaded medical documents.
- Apply access controls to claim information.
- Avoid displaying sensitive information to unauthorized users.

6.4 Reliability

The system shall maintain consistency between claims, policies, users, documents, assessments, and settlement information.

A successfully stored claim shall remain associated with the correct policyholder and policy.

6.5 Availability

The system should remain available during its intended operating period except during planned maintenance or unexpected infrastructure failures.

6.6 Usability

The system shall provide:

- Clear navigation
- Understandable labels
- Meaningful error messages
- Consistent layouts
- Simple claim submission
- Clear claim status information
- Responsive interfaces where applicable

6.7 Scalability

The system should be designed so that the number of policyholders, policies, claims, and documents can grow without requiring fundamental changes to the application's architecture.

6.8 Maintainability

The system should follow modular development principles.

The implementation should support:

- Separation of responsibilities
- Meaningful naming
- Reusable components
- Error handling
- Logging
- Configuration management
- Documentation

6.9 Portability

The web application should be accessible through commonly supported modern web browsers.

The system should not depend unnecessarily on a single client platform.

---

7. System Workflow

7.1 Overall Claim Lifecycle

The proposed claim lifecycle is:

Policyholder Login

↓

Select Policy

↓

Create Claim

↓

Enter Claim Information

↓

Upload Supporting Documents

↓

Submit Claim

↓

Claim Verification

↓

Additional Documents Required?

- Yes → Request Documents → Policyholder Uploads Documents
- No → Continue

↓

Claim Assessment

↓

Eligible?

- No → Reject Claim → Notify Policyholder
- Yes → Continue

↓

Claim Approval

↓

Settlement Processing

↓

Claim Closed

---

8. System Models and Diagrams

The final project documentation should contain the following diagrams.

8.1 Use Case Diagram

Primary actors:

- Policyholder
- Claim Officer
- Claim Assessor
- Administrator

Major use cases:

- Register
- Login
- View Policy
- Submit Claim
- Upload Documents
- Track Claim
- Verify Claim
- Assess Claim
- Approve Claim
- Reject Claim
- Process Settlement
- Manage Users
- Manage Policies
- Generate Reports

8.2 Activity Diagram

The primary activity diagram should model the complete claim submission and processing lifecycle.

It should include decision points for:

- Valid/invalid claim data
- Valid/invalid policy
- Missing documents
- Eligibility
- Approval/rejection

8.3 Sequence Diagrams

Recommended sequence diagrams:

1. User Registration
2. User Login
3. Claim Submission
4. Document Upload
5. Claim Verification
6. Claim Approval
7. Claim Rejection
8. Claim Settlement

8.4 ER Diagram

The ER diagram should represent the actual database entities and relationships.

Potential entities include:

- User
- Policyholder
- Employee
- Policy
- Claim
- Document
- Assessment
- Settlement
- Notification
- AuditLog

The final ER diagram must be synchronized with the actual database implementation.

8.5 Class Diagram

The class diagram should represent the major application components and domain entities.

Potential classes:

- User

- Policyholder
- Employee
- Administrator
- Policy
- Claim
- Document
- Assessment
- Settlement
- Notification
- AuditLog

8.6 Data Flow Diagram

The DFD should represent:

Policyholder → Portal → Claim Processing System → Database

and:

Claim Officer → Portal → Claim Processing → Database

and:

Administrator → Portal → Administration Module → Database

8.7 System Architecture

The recommended high-level architecture is:

Presentation Layer

↓

Application/API Layer

↓

Business Logic Layer

↓

Data Access Layer

↓

Database / Document Storage

External services, where applicable, may connect to the application layer.

---

9. Data Requirements

9.1 User Data

Potential attributes:

- User ID
- Name
- Email
- Contact information
- Password/authentication information
- Role
- Account status
- Creation date

9.2 Policy Data

Potential attributes:

- Policy ID
- Policy number
- Policyholder ID
- Policy type
- Coverage information
- Start date
- End date
- Policy status

9.3 Claim Data

Potential attributes:

- Claim ID
- Policy ID
- Policyholder ID
- Claim type
- Treatment details

- Hospital information
- Admission date
- Discharge date
- Claimed amount
- Approved amount
- Status
- Submission date
- Decision date

9.4 Document Data

Potential attributes:

- Document ID
- Claim ID
- Document type
- File name
- File location
- Upload date
- Verification status

9.5 Assessment Data

Potential attributes:

- Assessment ID
- Claim ID
- Assessor ID
- Assessment result
- Remarks
- Assessment date

9.6 Settlement Data

Potential attributes:

- Settlement ID

- Claim ID
- Approved amount
- Settlement status
- Settlement date
- Remarks

9.7 Notification Data

Potential attributes:

- Notification ID
- User ID
- Claim ID
- Notification type
- Message
- Created date
- Read status

9.8 Audit Data

Potential attributes:

- Audit ID
- User ID
- Action
- Entity affected
- Entity ID
- Timestamp
- Relevant description

---

10. Business Rules

The following business rules may be applied by the system.

BR-001

A user must be authenticated before accessing protected claim information.

BR-002

A policyholder may submit claims only for policies associated with their account.

BR-003

A claim must contain all mandatory information before submission.

BR-004

A claim must have a unique claim identifier.

BR-005

Required supporting documents must be submitted before the claim can proceed to the appropriate verification stage.

BR-006

Only authorized personnel may verify or assess claims.

BR-007

Only authorized personnel may approve or reject claims.

BR-008

A rejected claim must contain a rejection reason.

BR-009

A settlement record shall be associated with the corresponding approved claim.

BR-010

Users shall not be allowed to access claims for which they lack authorization.

BR-011

A claim shall follow the defined processing lifecycle.

BR-012

Significant claim-processing actions should be recorded for audit purposes.

---

11. Claim Status Model

The system should maintain clearly defined claim statuses.

A possible status model is:

Status	Description
Draft	Claim has been started but not submitted
Submitted	Claim has been submitted
Under Verification	Claim is being verified
Documents Required	Additional documents are required

Status	Description
Verified	Initial verification completed
Under Assessment	Claim is being assessed
Approved	Claim has been approved
Rejected	Claim has been rejected
Settlement Pending	Approved claim awaits settlement
Settled	Settlement has been recorded
Closed	Claim processing is complete

The exact statuses should be adjusted to match the implemented workflow.

---

12. Security and Access Control

12.1 Role-Based Access Control

Access shall be based on user roles.

Function	Policyholder	Claim Officer	Assessor	Admin
View Own Profile	✓	✓	✓	✓
Submit Claim	✓	—	—	✓*
View Own Claims	✓	—	—	✓
Review Claims	—	✓	✓	✓
Verify Claims	—	✓	✓	✓
Assess Claims	—	Limited	✓	✓
Approve Claims	—	As authorized	✓	✓
Reject Claims	—	As authorized	✓	✓
Manage Users	—	—	—	✓
Generate Reports	—	Limited	Limited	✓

* Administrative claim creation should only be included if supported by the project.

---

13. Error Handling and Exception Requirements

The system shall provide appropriate responses for common errors.

13.1 Authentication Errors

Examples:

- Invalid username/email
- Invalid password
- Missing credentials
- Unauthorized access

13.2 Claim Errors

Examples:

- Missing required fields
- Invalid policy
- Expired policy
- Invalid claim amount
- Duplicate claim
- Missing documents

13.3 Document Errors

Examples:

- Unsupported file format
- File too large
- Failed upload
- Missing required document

13.4 System Errors

Examples:

- Database unavailable
- Network failure
- Internal server error
- Resource unavailable

The system should display user-friendly messages without exposing sensitive internal implementation details.

---

14. Acceptance Criteria

A major feature shall be considered acceptable when its defined requirements have been implemented and verified.

Authentication

- Valid users can log in.
- Invalid credentials are rejected.
- Unauthorized users cannot access protected resources.
- Users receive functionality appropriate to their role.

Claim Submission

- An eligible policyholder can create a claim.
- Mandatory fields are validated.
- Required documents can be uploaded.
- A unique claim ID is generated.
- The submitted claim appears in the user's claim history.

Claim Verification

- Authorized personnel can access claims assigned or available to them.
- Supporting documents can be reviewed.
- Verification status can be updated.
- Additional documents can be requested.

Claim Assessment

- Authorized assessors can review verified claims.
- Assessment decisions can be recorded.
- Relevant remarks can be stored.

Approval/Rejection

- Authorized personnel can approve eligible claims.
- Authorized personnel can reject claims.
- Rejected claims contain a reason.
- Claim status is updated correctly.

Settlement

- Approved settlement information can be recorded.
- Settlement status can be updated.
- Settled claims can be closed.

Security

- Unauthorized users cannot access restricted functionality.
- Sensitive claim information is protected.
- Documents cannot be accessed without authorization.

---

15. Testing Requirements

The system shall be tested at multiple levels.

15.1 Functional Testing

Test:

- Registration
- Login
- Policy retrieval
- Claim submission
- Document upload
- Claim verification
- Assessment
- Approval
- Rejection
- Settlement
- Claim tracking

15.2 Security Testing

Test:

- Authentication
- Authorization
- Role restrictions
- Unauthorized API access
- Invalid tokens/session handling
- Input validation
- File upload validation

15.3 Database Testing

Test:

- Data creation
- Data retrieval
- Data modification
- Referential relationships
- Data consistency
- Duplicate prevention

15.4 Performance Testing

Test:

- Login response
- Claim search
- Claim submission
- Document upload
- Dashboard loading
- Concurrent users

15.5 Usability Testing

Test:

- Navigation
- Form usability
- Error messages
- Claim status visibility
- Responsive behavior

---

16. Requirements Traceability Matrix

Requirement ID	Module	Related Use Case	Verification
FR-AUTH-001	Authentication	UC-01 Registration	Functional Test
FR-AUTH-002	Authentication	UC-02 Login	Functional Test
FR-CLM-001	Claims	UC-03 Submit Claim	Functional Test

Requirement ID	Module	Related Use Case	Verification
FR-DOC-001	Documents	UC-04 Upload Document	Functional Test
FR-VER-001	Verification	UC-05 Verify Claim	Functional Test
FR-ASS-001	Assessment	UC-06 Assess Claim	Functional Test
FR-APP-001	Approval	UC-07 Approve Claim	Functional Test
FR-REJ-001	Rejection	UC-08 Reject Claim	Functional Test
FR-SET-001	Settlement	UC-09 Settle Claim	Functional Test
FR-TRK-001	Tracking	UC-10 Track Claim	Functional Test
NFR-SEC-001	Security	All Protected Functions	Security Test
NFR-PERF-001	Performance	System-wide	Performance Test

17. System Constraints

Potential constraints include:

1. The system requires network connectivity.
2. Users require a supported web browser.
3. Uploaded documents must comply with configured file restrictions.
4. System performance depends on the deployed server and database infrastructure.
5. External services may impose availability or usage limitations.
6. The system is limited by available storage capacity.
7. Actual insurance settlement may require integration with external financial systems if implemented in a production environment.

---

18. Assumptions and Dependencies

18.1 Assumptions

The system assumes that:

- Users provide valid information.
- Policy information available to the system is accurate.
- Authorized employees process claims according to defined procedures.
- Users have access to the internet.
- Users have access to a compatible web browser.

- Uploaded documents are genuine and relevant to the claim.

18.2 Dependencies

The system may depend on:

- Database availability
- Server availability
- Document storage
- Authentication mechanisms
- Notification services
- External APIs, if integrated

---

19. Limitations

Potential limitations of the current project include:

- The system may operate as a prototype rather than a production insurance platform.
- Real-world insurance-provider integration may not be available.
- Real banking/payment processing may not be implemented.
- Automated medical verification may not be available.
- Advanced fraud detection may not be implemented.
- Large-scale production performance may not have been evaluated.
- Actual legal/regulatory compliance may require additional controls beyond the academic implementation.

These limitations should be updated according to the actual implemented system.

---

20. Future Enhancements

Potential future enhancements include:

20.1 Mobile Application

Development of Android and iOS applications for policyholders and insurance employees.

20.2 Automated Claim Assessment

Use of machine learning and rule-based systems to assist with claim assessment.

20.3 Fraud Detection

Implementation of automated fraud detection based on claim patterns and historical data.

20.4 OCR-Based Document Processing

Use Optical Character Recognition to extract information automatically from uploaded medical documents.

20.5 Hospital Integration

Integration with hospitals and healthcare providers for direct submission and verification of medical information.

20.6 Payment Gateway Integration

Integration with secure payment systems for automated settlement processing.

20.7 Advanced Analytics

Dashboards showing:

- Number of claims
- Approved claims
- Rejected claims
- Average processing time
- Settlement amounts
- Claim trends

20.8 AI-Assisted Claim Processing

AI could assist authorized personnel in:

- Document classification
- Information extraction
- Duplicate detection
- Risk identification
- Claim prioritization

AI should remain an assistance mechanism rather than replacing authorized human decisions where human review is required.

---

21. Conclusion

To summarize, the Health Insurance Claim Portal tries to bring a fairly manual, paperwork-heavy process into a single structured system.

Right from claim submission and document upload to verification, assessment, the approve/reject decision, settlement, and finally tracking - each of these stages has been covered as part of this SRS.

Along with the functional side, this document also lays down the non-functional, security, data, interface, and workflow requirements that would be needed to actually build and evaluate the system.

If implemented as described, the system should make the whole claim process more transparent, cut down on manual back-and-forth, and give both policyholders and insurance staff a clearer, more structured way of tracking and handling claims.

---

22. Appendices

Appendix A — Recommended Use Cases

ID	Use Case
UC-01	Register Account
UC-02	Login
UC-03	View Policy
UC-04	Submit Claim
UC-05	Upload Documents
UC-06	Track Claim
UC-07	Verify Claim
UC-08	Request Additional Documents
UC-09	Assess Claim
UC-10	Approve Claim
UC-11	Reject Claim
UC-12	Process Settlement
UC-13	Manage Users
UC-14	Manage Policies
UC-15	Generate Reports

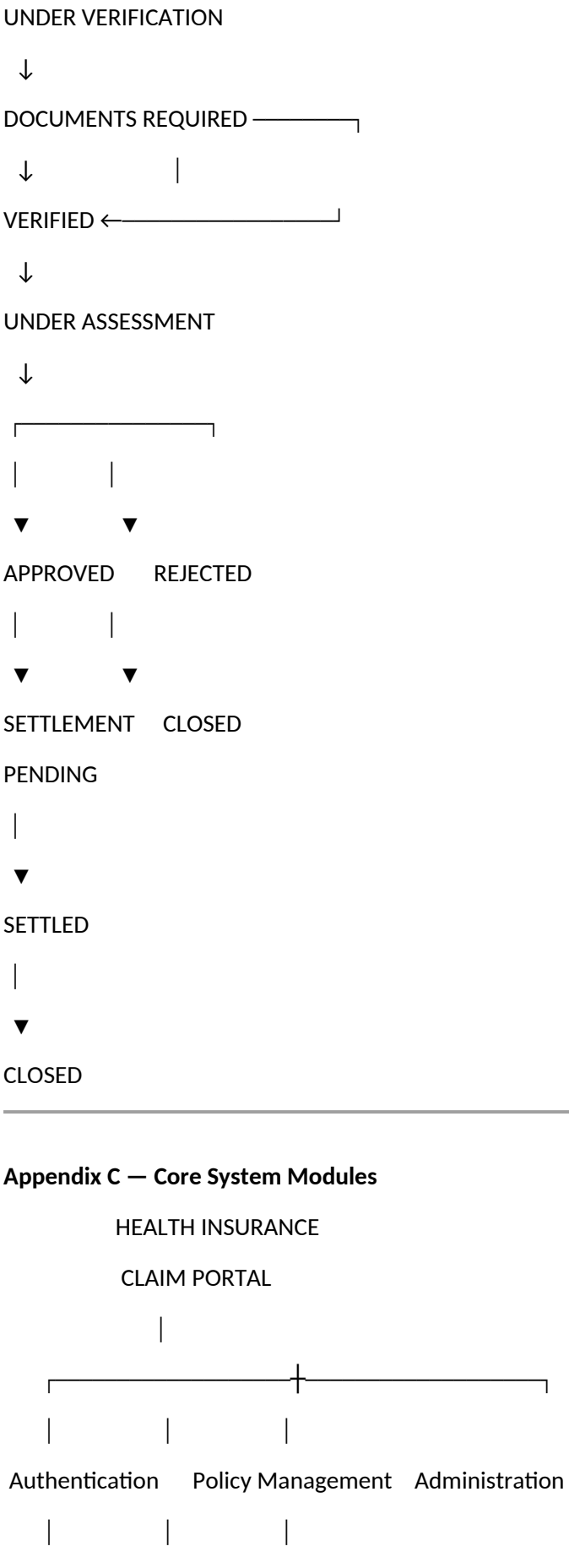
Appendix B — Recommended Claim Status Flow

DRAFT

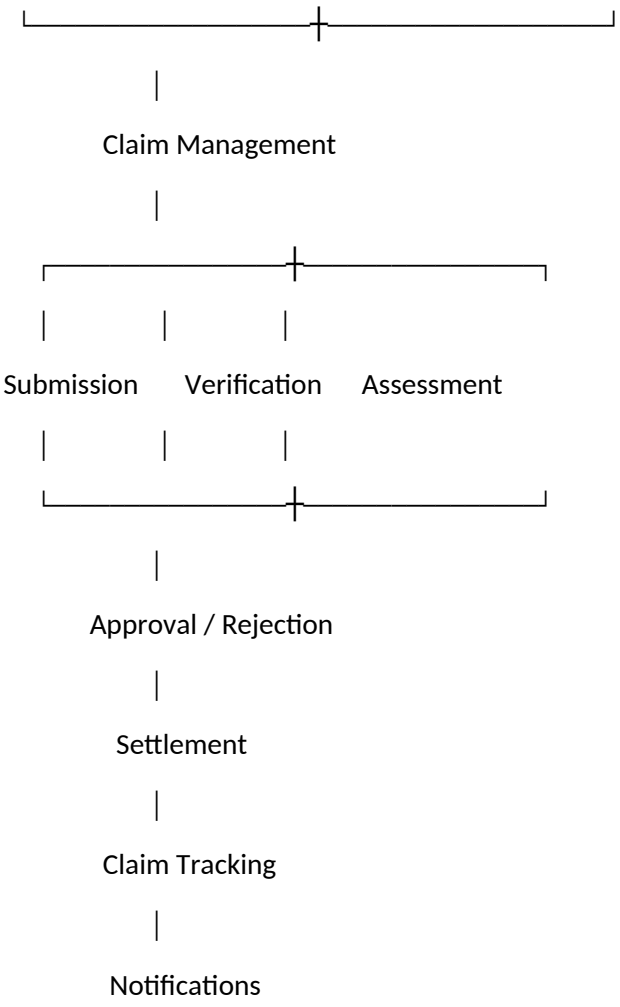
↓

SUBMITTED

↓



Appendix C — Core System Modules



Appendix D — Recommended Documentation Set

For the complete project submission, the SRS should ideally be accompanied by:

1. Use Case Diagram
2. Use Case Descriptions
3. ER Diagram
4. Class Diagram
5. DFD Level 0
6. DFD Level 1
7. Activity Diagram
8. Sequence Diagrams
9. System Architecture Diagram
10. Database Schema
11. API Documentation

12. Test Cases
13. Requirements Traceability Matrix
14. UI Screenshots
15. Deployment Architecture
16. Project User Manual